



Storm Intensity Distribution — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 06-April-2026

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1 Test Results — Storm Intensity Distribution (MKM-SI-001)

Tests run on **06 April 2026 at 07:49:50**.

Table 1: Test Results Summary — Storm Intensity Distribution (MKM-SI-001)

Total Tests	63
Passed	63
Failed	0
Skipped	0
Duration	0.02s

Table 2: Detailed Test Results — Storm Intensity Distribution

Test Description	Acceptance Criteria	Result	Time
All categories defined	<code>set(DURATION_PARAMS.keys()) == expected</code>	Pass	0.000s
Catastrophic extended	<code>high == 240</code>	Pass	0.000s
Each tier's max roughly doubles the previous.	<code>1.5 <= ratio <= 3.0</code>	Pass	0.000s
Extreme extended	<code>DURATION_PARAMS["extreme"] == (36, 72, 144)</code>	Pass	0.000s
Moderate extended	<code>low == 12; mode == 30 (+1 more)</code>	Pass	0.000s
Severe extended	<code>DURATION_PARAMS["severe"] == (24, 48, 96)</code>	Pass	0.000s
Gap types	<code>GapType.SHORT.value == "short";</code> <code>GapType.MEDIUM.value == "medium" (+1 more)</code>	Pass	0.000s
Sequence gap mapping	<code>SEQUENCE_GAP_TYPE[SequenceType.DOUBLET] == GapType.MEDIUM;</code> <code>SEQUENCE_GAP_TYPE[SequenceType.CLUSTER] == GapType.MEDIUM (+2 more)</code>	Pass	0.000s
Sequence types	<code>SequenceType.ISOLATED.value == "isolated";</code> <code>SequenceType.DOUBLET.value == "doublet" (+2 more)</code>	Pass	0.000s
Storm count ranges	<code>SEQUENCE_TYPE.STORM_COUNTS[SequenceType.ISOLATED] == (1, 1);</code> <code>SEQUENCE_TYPE.STORM_COUNTS[SequenceType.DOUBLET] == (2, 2) (+2 more)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Constants	EVENT_WINDOW_HOURS == 168; MIN_DRAINAGE_WINDOW_HOURS == 12 (+1 more)	Pass	0.000s
Exact boundary	fits_in_window([78.0, 78.0], [0.0])	Pass	0.000s
Exceeds window	not fits_in_window([72.0, 72.0], [48.0])	Pass	0.000s
Simple doublet fits	fits_in_window([24.0, 24.0], [48.0])	Pass	0.000s
Single storm	fits_in_window([100.0], [])	Pass	0.000s
Long gaps	GAP_PARAMS[GapType.LONG] == (48, 96, 144)	Pass	0.000s
Medium gaps	GAP_PARAMS[GapType.MEDIUM] == (24, 48, 72)	Pass	0.000s
Short gaps	GAP_PARAMS[GapType.SHORT] == (6, 18, 36)	Pass	0.000s
Doublet medium	get_gap_range(SequenceType.DOUBLET) == (24, 48, 72)	Pass	0.000s
Isolated zero	get_gap_range(SequenceType.ISOLATED) == (0, 0, 0)	Pass	0.000s
Get duration range	get_duration_range("moderate") == (12, 30, 60)	Pass	0.000s
Get max duration	get_max_duration("catastrophic") == 240; get_max_duration("minimal") == 12	Pass	0.000s
Ids unique	len(ids) == 100	Pass	0.000s
Sequence id format	sid.startswith("STORM-"); len(sid) == 14	Pass	0.000s
Sequence ids unique	len(ids) == 100	Pass	0.000s
Storm id format	sid.startswith("STORM-"); len(sid) == 14	Pass	0.000s
Invalid category	—	Pass	0.000s
Mode should be near the most frequent value in a large sample.	abs(median - mode) < 15	Pass	0.005s
Line 50: rng=None → creates np.random.RandomState() internally.	low <= d <= high	Pass	0.000s
Reproducibility	d1 == d2	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='baseline']	low <= d <= high	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='catastrophic']	low <= d <= high	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All sampled durations fall within [min, max] for the category. [category='extreme']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='minimal']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='moderate']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='severe']	<code>low <= d <= high</code>	Pass	0.000s
Doublet gaps should fall in medium range (24-72h).	<code>all(24 <= g <= 72 for g in gaps)</code>	Pass	0.000s
Isolated returns zero	<code>sample_gap(SequenceType.ISOLATED) == 0.0</code>	Pass	0.000s
Line 49: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.	<code>low <= g <= high</code>	Pass	0.000s
Persistent gaps should fall in short range (6-36h).	<code>all(6 <= g <= 36 for g in gaps)</code>	Pass	0.000s
Reproducibility	<code>g1 == g2</code>	Pass	0.000s
Within <code>bounds[sequencetype.cluster]</code> <code>[seq_type=i;SequenceType.CLUSTER:'cluster']</code>	<code>low <= g <= high</code>	Pass	0.000s
Within <code>bounds[sequencetype.doublet]</code> <code>[seq_type=i;SequenceType.DOUBLET:'doublet']</code>	<code>low <= g <= high</code>	Pass	0.000s
Within <code>bounds[sequencetype.persistent]</code> <code>[seq_type=i;SequenceType.PERSISTENT:'persistent']</code>	<code>low <= g <= high</code>	Pass	0.000s
Create basic	<code>storm.storm_id == "STORM-abc12345";</code> <code>storm.duration_hours == 24.0</code>	Pass	0.000s
From dict round trip	<code>restored.storm_id ==</code> <code>original.storm_id;</code> <code>restored.duration_hours ==</code> <code>original.duration_hours (+6 more)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Precipitation rounding	d["precipitation_mm"] == 35.12; d["intensity_factor"] == 1.234568 (+1 more)	Pass	0.000s
To dict	d["storm_id"] == "STORM-test1234"; d["start_time.hours"] == 48.0 (+6 more)	Pass	0.000s
Antecedent defaults	seq.antecedent_soil_moisture == "normal"; seq.antecedent_groundwater == "normal"	Pass	0.000s
Create doublet	seq.num_storms == 2; len(seq.storms) == 2 (+2 more)	Pass	0.000s
From dict round trip	restored.storm_id == original.storm_id; restored.duration_hours == original.duration_hours (+6 more)	Pass	0.000s
Isolated no gaps	len(seq.inter_storm_gaps_hours) == 0	Pass	0.000s
Spatial correlation defaults	seq.spatial_correlation_enabled is False; seq.spatial_correlation_range_km is None	Pass	0.000s
Spatial correlation enabled	d["spatial_correlation_enabled"] is True; d["spatial_correlation_range_km"] == 43.2	Pass	0.000s
To dict	d["storm_id"] == "STORM-test1234"; d["start_time.hours"] == 48.0 (+6 more)	Pass	0.000s
Duration mismatch	any("total_duration.hours" in e for e in errors)	Pass	0.000s
Exceeds precip window	any("MAX_PRECIP_END_HOUR" in e for e in errors)	Pass	0.000s
Invalid intensity category	any("invalid intensity" in e for e in errors)	Pass	0.000s
Invalid sequence type	any("Invalid sequence_type" in e for e in errors)	Pass	0.000s
Overlapping storms	any("starts at" in e and "before storm" in e for e in errors)	Pass	0.000s
Storm count mismatch	any("num_storms" in e for e in errors)	Pass	0.000s
Too many storms	any("not in [1, 5]" in e for e in errors)	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Valid doublet	errors == []	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (12 tests)	D. Kelly
07-Mar-2026	1.1	73 test(s) added; 12 test(s) removed (total: 146)	D. Kelly
08-Mar-2026	1.2	42 test(s) added (total: 115)	D. Kelly
12-Mar-2026	1.3	2 test(s) added (total: 117)	D. Kelly
27-Mar-2026	1.4	12 test(s) removed (total: 105)	D. Kelly
06-Apr-2026	1.5	42 test(s) removed (total: 63)	D. Kelly